Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/02/2020
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a		

Task Description:	representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with five embedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	Report covers work through December 31, 2019 (Ed., 12/2/2020) Fabrications of test samples of PMMA spheres for the space experiment are almost complete. The positions of 5 interior thermocouples were confirmed by X-ray scan for each sample. Fifteen more samples for additional ground tests are being prepared. This is a large effort because the involvement of several parties. A modeling paper simulating the flight experiment of flame growth and extinction has been published (Endo, M., T'ien, J.S., Ferkul, P.V., Olson, S.L., Johnston, M.C., Flame Growth Around a Spherical Solid Fuel in Low Speed Forced Flow in Microgravity, Fire Technology (2019) https://); see also Bibliography section.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Endo M, T'ien JS, Ferkul PV, Olson SL, Johnston MC. "Flame growth around a spherical solid fuel in low speed forced flow in microgravity." Fire Technology. 2020 Jan;56(1):5-32. First Online: 23 April 2019. https://doi.org/10.1007/s10694-019-00848-2 , Jan-2020